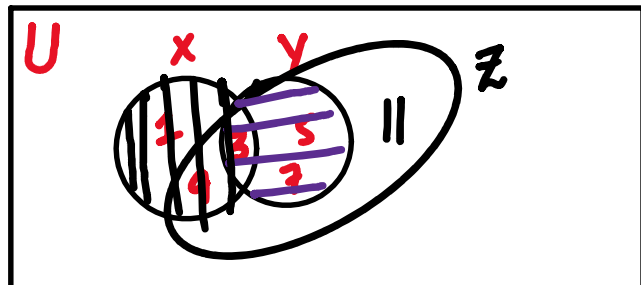


Math Class 9th Federal Board Chapter 03 Sets and Relations Venn Diagrams Exam Problems

- (v) If $X = \{1, 3, 9\}$, $Y = \{3, 5, 7\}$ and $Z = \{3, 5, 7, 9, 11\}$, then using a Venn diagram prove that $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$.

L.H.S =



$Y \cap Z$

$X \cup (Y \cap Z)$

$\equiv$

$\equiv, |||$

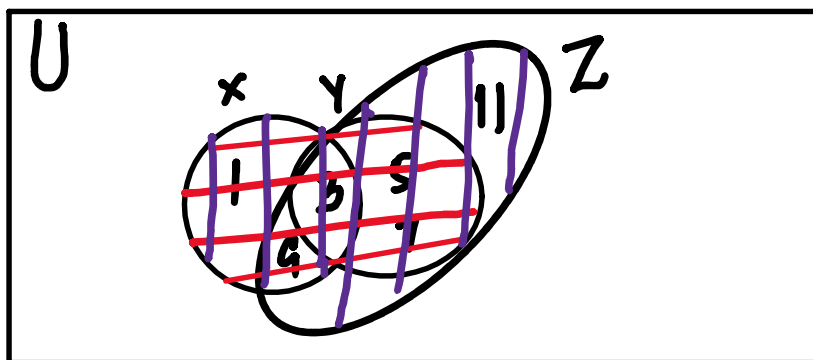
$$L.H.S. = X \cup (Y \cap Z) = \{1, 3, 5, 7, 9\}$$

$$R.H.S. = (X \cup Y) \cap (X \cup Z)$$

$$= \{1, 3, 5, 7, 9\} \cap \{1, 3, 5, 7, 9, 11\}$$

$$= \{1, 3, 5, 7, 9\}$$

R.H.S.



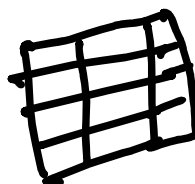
$X \cup Y$

$X \cup Z$

$(X \cup Y) \cap (X \cup Z)$

$\equiv$

$|||$

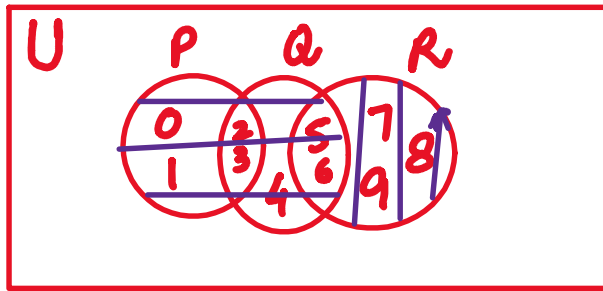


- (iii) Using a Venn diagram, prove that $(P \cup Q) \cup R = P \cup (Q \cup R)$, where $P = \{0, 1, 2, 3\}$, $Q = \{2, 3, 4, 5, 6\}$ and $R = \{5, 6, 7, 8, 9\}$.

L.H.S =

3, 4, 5, 6} and $R = \{5, 6, 7, 8, 9\}$.

L.H.S. =



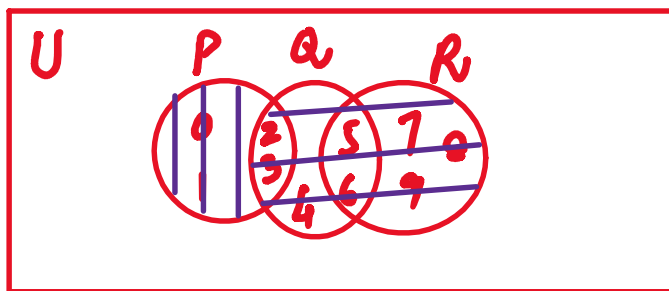
$P \cup Q$

$\equiv$

$(P \cup Q) \cup R$

$\equiv, |||$

R.H.S. =



$Q \cup R$

$\equiv$

$P \cup (Q \cup R)$

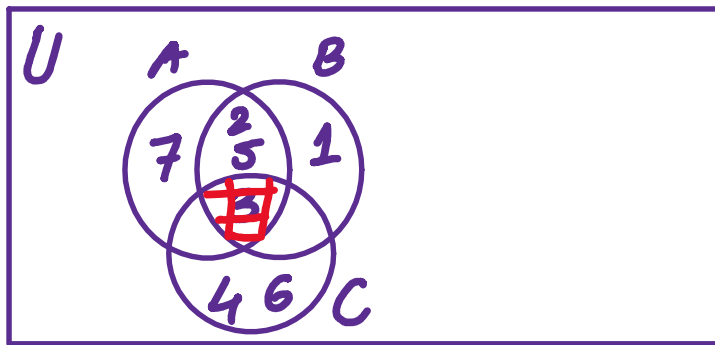
$\equiv, |||$

Hence, L.H.S. = R.H.S.
Thus, proved.

(vii) Verify using a Venn diagram that $A \cap (B \cap C) = (A \cap B) \cap C$ for $A = \{2, 3, 5, 7\}$, $B = \{1, 2, 3, 5\}$ and $C = \{3, 4, 6\}$.

L.H.S. =

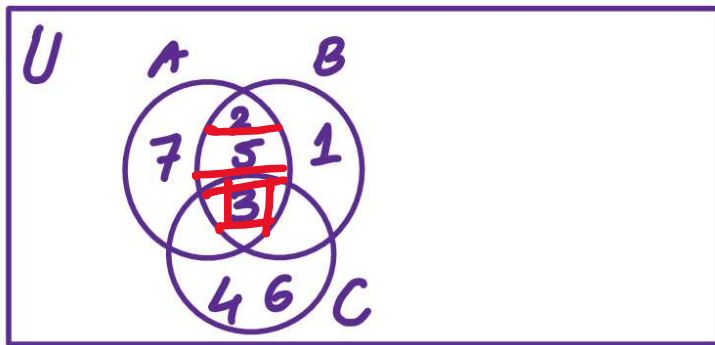




$$B \cap C \quad \equiv$$

$$A \cap (B \cap C) \quad \equiv \#$$

R.H.S =



$$A \cap B \quad \equiv$$

$$(A \cap B) \cap C \quad \equiv \#$$

Thus, L.H.S = R.H.S.

Hence proved.

$$\{1, 2\} \cup \{3\} = \{1, 2, 3\}$$

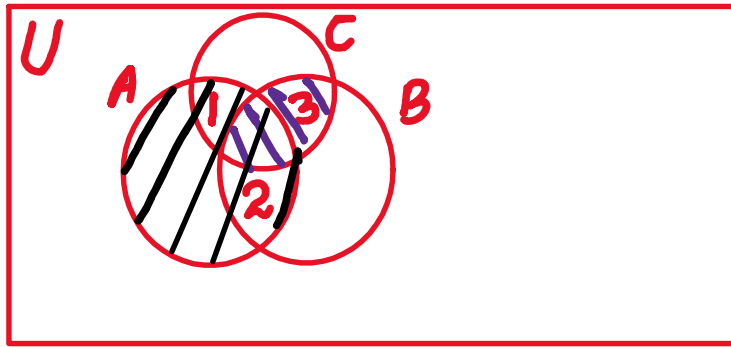
- (x) Use a Venn diagram to verify that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, where $A = \{1, 2\}$, $B = \{2, 3\}$ and $C = \{1, 3\}$.

$$\{1, 2, 3\} \cap \{1, 2, 3\} = \{1, 2, 3\}$$

3} and $C = \{1, 3\}$.

$$\{1, 2, 3\} \cap \{1, 2, 3\} = \{1, 2, 3\}$$

L.H.S =



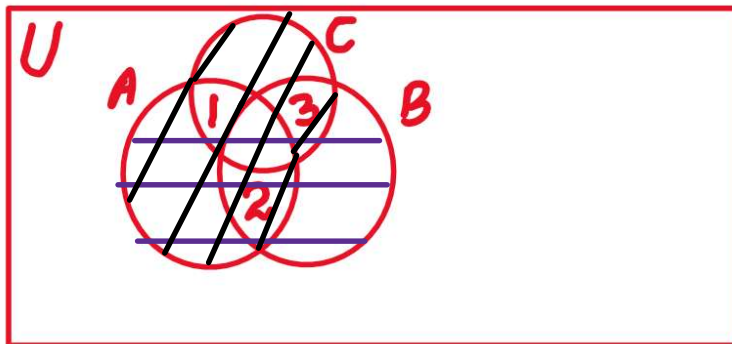
$B \cap C$

///

$A \cup (B \cap C)$

///, ///

R.H.S



$A \cup B$

==

$A \cup C$

///

$(A \cup B) \cap (A \cup C)$

###

Hence. L.H.S = R.H.S.
Proved

Thus, prover